

Absolute error



1

a 0.5 kg

b 0.5 g

c \$5

2 $\frac{1}{2}$ mL

3

a 0.05 mL

b 0.5 mm

c 0.05 g

d 0.5 s

e 0.005 kg

f 0.05 m

g 0.005 °C

4 We do not know what place value the number has been rounded to.

The limits of accuracy

5

a 1 cm

b 90 cm

c 0.5 cm

d

i No

ii No

iii No

iv Yes

6

a 10 kg

b 5 kg

c Lower bound: 65 kg, upper bound: 75 kg

7

a 50 km/h

b 5 km/h

c 55 km/h

8

a

i 13.455 km

ii 13.445 km

b

i 6.45 km

ii 6.35 km

c

i 5.5 m

ii 4.5 m

d

i 14.005 kg

ii 13.995 kg

e

i 4884.5 m

ii 4883.5 m

f

i \$52.50

ii \$47.50

9 19.995 m

10 2.05 m

11

a Maximum: 175, minimum: 165



b Maximum: 3050, minimum: 2950

Further calculations (Extended)

12

a 26.25

b 0.6

13

a 13.85 m/s

b 13.95 m/s

14

a 153.9 cm^2

b $132.7 \text{ cm}^2, 176.7 \text{ cm}^2$

c 22.8 cm^2

d 14.8%

15

a Upper bound: 272.6925 m^2 , lower bound: 269.3925 m^2

b Upper bound: 66.2 m, lower bound: 65.8 m

16

a Lower bound: 9.5 cm, upper bound: 10.5 cm

b 100 cm^2

c Lower bound: 90.25 cm^2 , upper bound: 110.25 cm^2

d 1000 cm^3

e Lower bound: 857.375 cm^3 , upper bound: 1157.625 cm^3